# Exchange landscapes, orientation, and rigidity in the Golden six-mode conference interferometer

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**Abstract**

Which features of a one-particle transfer remain observable when its ports are unframed, oriented, or phase-calibrated, and how do those features vary under continuous control and conference phase? Singular values classify an unframed real transfer. Orienting the ports adds the sign of the top exterior amplitude; up to scale, the determinant is the unique minimal-degree polynomial carrying this relative orientation. Fully calibrated frames expose matrix-dependent amplitudes such as the permanent.

For symmetric conference matrices, the balanced exchange spectrum is a normalized cross-Gram spectrum; order six is the unique nontrivial order at which it is independent of the balanced cut. In the Golden order-six transfer, the balanced Boolean controls are precisely the joint maximizers of all degree-three Schur sectors over the real control cube. Hermitian conference phase gives a transverse deformation. Triangle holonomy fixes the first two exchange moments and moves the degree-three sectors along an exact Pareto segment. Constancy of any one sector characterizes the real switching class. An averaged squared-holonomy defect controls the Frobenius distance to the real conference orbit globally from below and locally from above.

The Golden realization retains a calibrated determinant-sign code and a photonic design boundary. Common-reference tomography and ordinary-photon controls test its one-particle carrier and bosonic shadow, whereas direct three-fermion emulation needs an additional antisymmetric three-qutrit source. This is a theory and design-limit analysis, not a report of a built device.

## 1 Introduction

Let $K$ be a real input–output block of a linear one-particle transfer. Without port frames, its singular values are intrinsic. Orienting the two port spaces turns the top exterior map into the signed determinant. Fixing ordered phase-calibrated frames then exposes matrix-dependent amplitudes, including bosonic collision-free permanents. These are different levels of information, not three notations for the same observable; Figure 1 records the hierarchy.

We study a six-mode transfer obtained from a real symmetric conference matrix $C$ of order six, so that

$$C^{\mathsf{T}} = C, \qquad C_{ii} = 0, \qquad C_{ij} \in \{\pm 1\}, \qquad C^2 = 5I.$$

The two eigenspaces of $C$, with eigenvalues $\pm\sqrt{5}$, both have dimension three. A diagonal path control couples them through a three-by-three block $K$. Six coherently marked conference matrices produce six such blocks. The Clebsch source papers construct this Golden family from the regular six-point two-graph and identify its determinants with the six classical Joubert coordinates on the Segre cubic [1, 2]. We ask what this transfer retains under

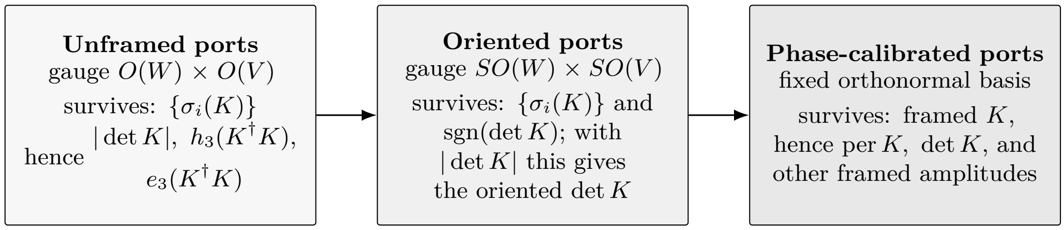


Figure 1: Observable hierarchy for a one-particle transfer $K : V \to W$. Each step fixes more port structure and exposes more information. At the unframed level every scalar invariant factors through the singular values. Orientations add the determinant sign off the singular locus; ordered phase-referenced frames then expose matrix-dependent amplitudes such as the permanent.

symmetric and antisymmetric exchange, how a calibrated apparatus reads its marked determinant signs, and where a photonic realization stops. The name *Golden* refers to the paired $\pm\sqrt{5}$ conference eigenspaces; *conference interferometer* refers to the operational realization.

Three reductions organize the argument. First, the left–right orbit theorem identifies what survives unframed and oriented port gauges and forces the determinant as the minimal orientation carrier. Second, a conference block identity turns balanced exchange spectra into cross-Gram spectra; it yields the all-order classification and supplies the endpoint data for the continuous Golden optimum. Third, triangle holonomy becomes the sole moving coordinate in the Hermitian order-six problem. A pentagon parity obstruction and a Pfaffian-square obstruction force the rigid endpoint, while sign rounding turns the same holonomy defect into a local metric estimate. These arguments use neither a classification of the Hermitian family nor a finite census.

The mechanism is the determinant line. The Golden data determine the two three-dimensional port spaces but not ordered orthonormal frames inside them. Changing those frames acts on $K$ from the left and right by orthogonal matrices. The top exterior power is intrinsically a map between the two determinant lines; after both lines are oriented, it becomes the signed scalar $\det K$, while the filled-fermion probability $|\det K|^2$ is sign-blind. The permanent does not descend through the same port-frame action. The canonical intrinsic bosonic aggregate is therefore a spectral quantity on the symmetric cube, not a second signed Segre coordinate.

The observable hierarchy has three experimental levels. Phase-referenced one-particle tomography infers calibrated determinant signs; ordinary photons measure bosonic controls; direct antisymmetric emulation needs an additional three-qutrit resource.

The permanent description of multiphoton scattering is standard [3]; immanants organize the intermediate symmetry sectors [4], and general partial-distinguishability formalisms precede the complementarity identities used here [5]. General boson–fermion complementarity in linear interferometers is due to Jabbour and Cerf [6]; its extension to partially distinguishable particles is due to Robbio, Jabbour, and Cerf [7].

The Clebsch source paper already records the determinant-line interpretation and the failure of the permanent to descend through port-frame changes [2, Section 5]. Triple products as switching invariants and ordinary three-uniform ETF compression spectra are established [8, 9, 10]; equal characteristic polynomials of Hermitian principal blocks form the distinct spectral-monomorphy problem [11]; uniform subframe bounds are another adjacent but unsquared condition [12]. The order-six Hermitian conference family used below is also prior [13]; a later work states an order-six classification [14]. Our rigidity concerns the sign-blind squared complementary cross-block spectrum rather than the ordinary principal spectrum. Here the orientation hierarchy, exchange landscape, and quantitative real-orbit stability meet in one transfer model with an explicit measurement boundary.

## 2 The Golden transfer

On the ordered path set $X = \{0, \ldots, 5\}$, take

$$C_0 = \begin{pmatrix} 0 & 1 & 1 & 1 & -1 & -1 \\ 1 & 0 & -1 & -1 & -1 & -1 \\ 1 & -1 & 0 & 1 & 1 & -1 \\ 1 & -1 & 1 & 0 & -1 & 1 \\ -1 & -1 & 1 & -1 & 0 & -1 \\ -1 & -1 & -1 & 1 & -1 & 0 \end{pmatrix}, \qquad C_0^2 = 5I.$$

Let $\mathcal{T}$ be the six synthematic totals, that is, the six labelled one-factorizations of $K_X$; below $ij$ denotes the edge $\{i, j\}$. One total is

$$T_0 = \{\{01, 23, 45\}, \{02, 14, 35\}, \{03, 15, 24\}, \{04, 13, 25\}, \{05, 12, 34\}\}.$$

The $S_6$-orbit of $T_0$ has six elements. For each $T \in \mathcal{T}$, choose the lexicographically first permutation $\pi_T$ carrying $T_0$ to $T$, let $P_T$ be its permutation matrix, and put $C_T = P_T C_0 P_T^{\mathsf{T}}$. We use this convention throughout. Transporting the ordered path basis fixes the Hodge and determinant-line orientations, including the signs $\varepsilon_T$ below. The Clebsch source theorem proves that this marked family is the conference-operator realization of the six Joubert coordinates [2, Theorem 5.1]; Paper I supplies its upstream two-graph provenance [1].

For one member $C_T$ of this marked conference family, let $Q_{T,+}$ and $Q_{T,-}$ be matrices whose columns are ordered orthonormal frames for the $\pm\sqrt{5}$-eigenspaces. In the base gauge these frames are obtained by Gram–Schmidt from $P_\pm e_0, P_\pm e_1, P_\pm e_2$, where $P_\pm = (I \pm C_0/\sqrt{5})/2$, and are transported by $P_T$. For $x = (x_0, \ldots, x_5)$, put

$$D_x = \operatorname{diag}(x_0, \ldots, x_5), \qquad K_T(x) = Q_{T,-}^{\mathsf{T}} D_x Q_{T,+}.$$

When $|x_i| = 1$, the six-mode transfer

$$U_T(x) = [Q_{T,+}\ Q_{T,-}]^{\mathsf{T}} D_x [Q_{T,+}\ Q_{T,-}]$$

is orthogonal. Values $|x_i| < 1$ describe a diagonal filter followed by postselection.

The six determinant amplitudes satisfy

$$Z_T(x) = \varepsilon_T\, 10\sqrt{5} \det K_T(x), \qquad \varepsilon_T \in \{\pm 1\}, \tag{1}$$

with signs fixed by the marked orientations. They obey

$$\sum_T Z_T(x) = 0, \qquad \sum_T Z_T(x)^3 = 0. \tag{2}$$

These are the Segre equations in Joubert coordinates. The equations and the outer $S_6$-action are classical invariant theory [18]; here we use their realization as six amplitudes of one marked transfer.

The six coordinate lines obtained from either three-dimensional eigenspace form the classical real $\mathrm{ETF}(3, 6)$ associated with a symmetric conference matrix [19]. In measurement language these are the six icosahedral lines of the real three-level SIC [20]; they are not the nine-line informationally complete SIC for a complex qutrit. This distinction matters here because the transfer is real and its port orientation is part of the calibrated data.

## 3 Why the determinant retains orientation

The source construction identifies the determinant line and gives the basic plane-rotation counterexample for the permanent [2, Section 5]. We sharpen that observation by classifying the full and oriented double orbits in every dimension and proving that the determinant is the unique relative invariant of minimal degree.

Write $H = K^\dagger K$. For later exchange comparisons, define

$$B_{\mathrm{sym}}(K) = \mathrm{tr}(\mathrm{Sym}^3 H) = h_3(H), \qquad F_{\mathrm{ext}}(K) = \mathrm{tr}(\textstyle\bigwedge^3 H) = e_3(H) = \det H.$$

The dagger notation keeps the statement valid for the complex port-gauge extension; throughout the real Golden construction it is simply transpose. The first expression sums, over the ten normalized three-boson input occupations, the probabilities that all three bosons transfer. Dividing by ten gives the uniform-input average $B_{\mathrm{sym}}/10$. The second expression is the filled three-fermion probability.

For an orthogonal sign control, the singular values of $K$ are the cosines of the principal angles between the output space and the controlled image of the input space. For a filtered control they remain transfer singular values, but need not be principal-angle cosines. Their product is the absolute determinant. After orientations are chosen, its sign records whether the induced map between the determinant lines preserves or reverses orientation.

**Theorem 3.1** (Left–right orbit and orientation boundary)**.** *Let $V$ and $W$ be real Euclidean spaces of the same dimension $n \geq 2$, and let $K : V \to W$.*

(i) *A change of ordered orthonormal frames sends the matrix of $K$ to*

$$K \longmapsto R_-^{\mathsf{T}} K R_+, \qquad R_\pm \in O(n).$$

*The $O(W) \times O(V)$ orbit is determined by the singular values. If $K$ is invertible, this orbit splits into two $SO(W) \times SO(V)$ orbits, distinguished by the sign of the oriented determinant. If $K$ is singular, it does not split.*

(ii) *Intrinsically, $\bigwedge^n K$ is a map $\det V \to \det W$. Choosing orientations turns it into a scalar, and a frame change multiplies that scalar by $\det(R_-)\det(R_+)$. Thus its absolute value is an orthogonal invariant, while its sign is an invariant of oriented port spaces.*

(iii) *The permanent of a matrix does not descend to either the orthogonal or the special-orthogonal double orbit. Hence it is a calibrated port-frame amplitude rather than a scalar attached to $K$ as an unframed Euclidean map.*

*Proof.* The singular-value decomposition proves that two matrices lie in the same $O(W) \times O(V)$ orbit exactly when they have the same singular values. For an invertible map, special-orthogonal changes preserve the sign of the oriented determinant. Conversely, singular-value decompositions of two maps with the same singular values and determinant sign can be chosen with the same pair of orientation parities, so special-orthogonal left and right factors relate them. When a singular value is zero, reflecting its kernel direction changes one parity without changing the diagonal singular-value matrix. The two orientation classes therefore merge on the singular locus.

The functorial map on top exterior powers gives (ii), including its frame transformation law. For (iii), let $R_\theta$ be a plane rotation and use the special-orthogonal family $R_\theta \oplus I_{n-2}$. Along the right orbit of the identity matrix,

$$\mathrm{per}(R_\theta \oplus I_{n-2}) = \cos(2\theta), \qquad \det(R_\theta \oplus I_{n-2}) = 1.$$

The permanent therefore varies inside a single special-orthogonal double orbit. $\square$

**Proposition 3.2** (The determinant is the minimal orientation carrier)**.** *Up to scale,* $\det K$ *is the unique degree-$n$ polynomial in the entries of $K$ satisfying*

$$F(R_-^{\mathsf{T}} K R_+) = \det(R_-)\det(R_+)F(K) \qquad (R_\pm \in O(n)).$$

*In particular, an orientation-covariant amplitude of the lowest possible degree is forced to be the top exterior amplitude.*

*Proof.* Apply diagonal sign changes on the left. Every monomial of a degree-$n$ relative invariant must have odd degree in each of the $n$ rows, hence degree one in every row. Right sign changes give degree one in every column. The same parity argument shows that every nonzero relative invariant has degree at least $n$. The polynomial is therefore a linear combination of the $n!$ permutation monomials. Invariance under orientation-preserving rotations in each pair of rows, followed by a reflection in one row, forces the coefficients of permutations differing by a transposition to have opposite signs. All coefficients are thus one common scalar times the permutation sign, which is the determinant expansion. □

Every scalar invariant under the full orthogonal port action consequently factors through the singular values. This statement concerns the unframed map. A laboratory may freeze ordered phase-referenced port frames and then measure additional calibrated amplitudes.

**Corollary 3.3** (Orientation chambers)**.** *On the invertible real transfer space, the two oriented double orbits over a fixed singular-value point are exchanged by reversing one port orientation. They meet only on the determinantal hypersurface. Thus* $\det K = 0$ *is exactly the boundary at which the oriented top-exterior label disappears.*

*Remark* 3.4 (Complex port gauges). For complex port frames, $O(n)$ and $SO(n)$ are replaced by $U(n)$ and $SU(n)$. Singular values again classify the full double orbit. On the invertible locus, the residual label is the phase of the oriented determinant; on the singular locus it disappears. The real Golden transfer and a coherent phase reference reduce this determinant-line phase to the sign studied here.

# 4 Real conference exchange spectra

The six-mode benchmark is the exceptional endpoint of a general conference classification, not the first member of a uniform family. Let $C = C^{\mathsf{T}}$ be a zero-diagonal sign matrix of order $2d$ with $C^2 = qI$, where $q = 2d - 1$. For a balanced half $Y$, write

$$C = \begin{pmatrix} A & R \\ R^{\mathsf{T}} & E \end{pmatrix}, \qquad H_Y = Q_+^{\mathsf{T}} D_Y P_- D_Y Q_+,$$

where $P_\pm = (I \pm C/\sqrt{q})/2$, $Q_+$ is an isometry onto $\operatorname{im} P_+$, and $D_Y$ is the sign involution of the cut. Complementing the half gives $D_{Y^c} = -D_Y$ and hence $H_{Y^c} = H_Y$; all cut counts below are therefore projective.

**Theorem 4.1** (Conference exchange rigidity)**.** *For every symmetric conference matrix,*

$$\operatorname{spec}(H_Y) = \operatorname{spec}\left(\frac{1}{q} R R^{\mathsf{T}}\right) = \left\{1 - \frac{\alpha_i^2}{q} : 1 \le i \le d\right\},$$

*where the $\alpha_i$ are the eigenvalues of $A$. This spectrum is independent of the balanced half exactly when $d \le 3$. Hence order six is the unique nontrivial symmetric conference case* *with uniform balanced exchange spectrum. The order-two case is the trivial one-mode case, and no symmetric conference matrix of order four exists.*

*For a four-set $J$, let $w(J)$ be the sum of its three signed Hamilton-cycle products, call $J$* aligned *when $w(J) = 3$, and let $c_Y$ count aligned four-sets contained in $Y$. Then*

$$\operatorname{tr}(H_Y^2) = \frac{F_d + 32c_Y}{q^2}, \qquad F_d = dq^2 - 2qd(d-1) + d(d-1) + 12\binom{d}{3} - 8\binom{d}{4}.$$

*Hence the purity is cut-independent exactly in the same small orders.*

*Proof.* Put $Q = C/\sqrt{q}$ and $L = [D_Y, Q]$. The commutator anticommutes with $Q$, and on the positive eigenspace $Lv = 2P_- D_Y v$. In cut coordinates,

$$L = \frac{2}{\sqrt{q}} \begin{pmatrix} 0 & R \\ -R^\mathsf{T} & 0 \end{pmatrix}.$$

Thus $4H_Y$ is $L^*L$ on that eigenspace, and $H_Y$ has the squared singular values of $R/\sqrt{q}$. Since the principal block of $C^2 = qI$ is $RR^\mathsf{T} = qI - A^2$, the spectral formula follows.

The identities $\operatorname{tr}(A^2) = d(d-1)$ and

$$\operatorname{tr}(A^4) = d(d-1) + 12\binom{d}{3} + 8\sum_{\substack{J\subset Y\\ |J|=4}} w(J)$$

give the displayed purity formula, because $w(J)$ is 3 on aligned four-sets and $-1$ otherwise. Suppose $d \geq 4$ and this purity is independent of $Y$. The sums of the aligned-set indicator over all $d$-subsets are then constant. The characteristic-zero inclusion matrix from four-sets to $d$-sets has full column rank [17, Theorem 1], so the indicator is constant on four-sets.

Switch $C$ so that every edge incident with one vertex is positive. For a four-set containing that vertex, if the other three edge signs are $x, y, z$, then $w = xy + xz + yz$. If the common value is 3, every triangle on the other $2d-1$ vertices is monochromatic; all its edges then have one sign, contradicting orthogonality of two conference rows. If the common value is $-1$, the edge two-colouring has no monochromatic triangle, and $R(3,3) = 6$ gives $2d-1 \leq 5$, again a contradiction.

For $d = 1$ the claim is immediate, and for $d = 2$ every two-by-two principal sign block squares to $I$. For $d = 3$, if $\tau$ is the product of the three edge signs, then $A^2 = 2I + \tau A$, so $A^2$ always has spectrum $\{4, 1, 1\}$. This proves both directions. □

At higher orders only statistics remain universal. Greaves and Suda proved that the aligned sets form a 3-$(2d, 4, (d-3)/2)$ design [15, Example 2.3]. Classical Johnson inclusion calculus [16] gives, for a uniformly chosen balanced half and $d \geq 4$,

$$\mathbb{E}c_Y = \rho\binom{d}{4}, \qquad \operatorname{Var}(c_Y) = \frac{\binom{2d}{4}\binom{2d-8}{d-4}}{\binom{2d}{d}}\rho(1-\rho), \qquad \rho = \frac{d-3}{2(2d-3)}.$$

The affine formula above therefore gives switching-class-independent mean and variance for the purity; its possible values are quantized in steps of $32/q^2$. At order ten the aligned sets form a Steiner system and force the 126 projective cuts to split into 36 with $\operatorname{tr}(H_Y^2) = 145/81$ and 90 with $\operatorname{tr}(H_Y^2) = 59/27$. Paper III proves the same general theorem at the conference-operator stage [2, Theorem 5.1]; the local proof above keeps this observable claim independently assessable. The aligned design and Johnson calculus are classical; here they yield the exchange interpretation and rigidity classification.

*Remark* 4.2 (Collective inverse boundary). At order $2d \geq 10$, the complete labelled purity landscape recovers every aligned four-set by inclusion inversion. Paper III proves that this family determines the discrete signing up to switching and global negation [2, Theorem 5.2]; one calibrated triangle product selects the remaining bit relative to that calibration. This is a collective inverse corollary, not a local optical observable: the mean and variance alone are universal, and each individual $H_Y$ remains orientation-blind.

**Theorem 4.3** (Golden balanced exchange benchmark)**.** *Let $K_T(x)$ be a Golden transfer block. At every balanced control with three entries $+1$ and three entries $-1$, for each of the six marked protocols,*

$$\operatorname{spec} H = \left\{ \frac{1}{5}, \frac{4}{5}, \frac{4}{5} \right\}.$$

*Consequently*

$$B_{\text{sym}} = \frac{313}{125}, \qquad F_{\text{ext}} = \frac{16}{125}, \qquad B_{\text{sym}} - F_{\text{ext}} = \frac{297}{125}.$$

*The corresponding uniform average over the ten normalized symmetric-cube input occupations is $B_{\text{sym}}/10 = 313/1250$. The mixed exchange sector has*

$$\operatorname{tr}\big(S_{(2,1)}(H)\big) = s_{(2,1)}(H) = \frac{8}{5},$$

*and the Schur–Weyl checksum is*

$$h_3(H) + 2s_{(2,1)}(H) + e_3(H) = \operatorname{tr}(H)^3 = \frac{729}{125}.$$

*In particular, every scalar invariant under the full orthogonal port action has the same value on all these balanced blocks and cannot distinguish their marked Segre nodes or orientations.*

*Proof.* Order the coordinates by the negative support and its complement, and write

$$C_T = \begin{pmatrix} A & R \\ R^{\mathsf{T}} & E \end{pmatrix}.$$

If $\tau$ is the product of the three off-diagonal entries of $A$, then

$$A^2 = 2I + \tau A, \qquad \det(tI - A) = t^3 - 3t - 2\tau.$$

Thus $A^2$ has spectrum $\{4, 1, 1\}$. The cross-Gram identity $RR^{\mathsf{T}} = 5I - A^2$ and Theorem 4.1 give the displayed spectrum of $H$ for every balanced support, without a census. Equivalently,

$$\det(\lambda I - H) = \lambda^3 - \frac{9}{5}\lambda^2 + \frac{24}{25}\lambda - \frac{16}{125} = \left(\lambda - \frac{1}{5}\right)\left(\lambda - \frac{4}{5}\right)^2. \tag{3}$$

The complete and elementary symmetric functions then give

$$h_3\left(\frac{1}{5}, \frac{4}{5}, \frac{4}{5}\right) = \frac{313}{125}, \qquad e_3\left(\frac{1}{5}, \frac{4}{5}, \frac{4}{5}\right) = \frac{16}{125}.$$

Finally, the degree-three symmetric-function identity $h_3 - e_3 = p_1 p_2$ gives $297/125$. For the remaining partition of three,

$$s_{(2,1)} = \frac{p_1^3 - p_3}{3} = \frac{8}{5}.$$

Schur–Weyl decomposition

$$V^{\otimes 3} = \operatorname{Sym}^3 V \oplus 2S_{(2,1)} V \oplus \bigwedge^3 V$$

then gives the checksum. The final assertion follows from Theorem 3.1. $\square$

## 5 Continuous control and the Hermitian exchange landscape

The balanced Boolean point is not only a discrete benchmark. It is the exact joint optimum of the intrinsic degree-three exchange sectors over the entire real control cube.

Since $K_T(x) = Q_{T,-}^{\mathsf{T}} D_x Q_{T,+}$ is a compression of $D_x$, it is a contraction on this cube.

**Theorem 5.1** (Continuous Golden control)**.** *For the real Golden conference matrix and every $x \in [-1,1]^6$, let $H(x) = K_T(x)^{\mathsf{T}} K_T(x)$ and $p_j = \operatorname{tr}(H^j)$. Then*

$$p_1 \le \frac{9}{5}, \qquad p_2 \le \frac{33}{25}, \qquad e_2 \le \frac{24}{25}, \qquad e_3 \le \frac{16}{125},$$

*and*

$$h_3 \le \frac{313}{125}, \qquad s_{(2,1)} \le \frac{8}{5}.$$

*Equality in any one of the three degree-three Schur bounds $h_3, s_{(2,1)}, e_3$ occurs exactly at the twenty balanced Boolean controls. Thus those controls are the unique joint maximizers of every degree-three exchange sector on the cube.*

*Proof.* The projector diagonal gives

$$p_1 = \frac{6\sum_i x_i^2 - (\sum_i x_i)^2}{20} \le \frac{9}{5}.$$

The fourth Schatten power $p_2 = \|K\|_{S_4}^4$ is convex in the linear matrix $K(x)$, hence is maximized at a cube vertex. Up to complement, Boolean negative-support sizes $0, 1, 2, 3$ have squared-singular spectra

$$(0,0,0), \quad (1,0,0), \quad (4/5, 4/5, 0), \quad (4/5, 4/5, 1/5),$$

which gives $p_2 \le 33/25$. Every two-by-two minor of $K(x)$ is affine in each individual coordinate: the quadratic coefficient vanishes because one path contributes a rank-one matrix. Hence the sum of their squares, $e_2$, is separately convex and reduces to the same vertices. The determinant is multi-affine for the same reason, so its squared modulus gives $e_3 \le 16/125$, with equality exactly at the balanced vertices. Indeed, equality at an interior coordinate would force equality at both endpoints of that coordinate interval. Iterating would produce two maximizing Boolean vertices differing in one sign, whereas every maximizing Boolean vertex is balanced. Hence no non-Boolean equality case occurs.

Now $h_3 = e_3 + p_1 p_2$, which proves its bound and forces the same equality case. The mixed sector is the only bound not obtained directly from separate convexity. An elementary three-variable lemma says that if $0 \le \lambda_i \le 1$, $e_1 \le 9/5$, and $e_2 \le 24/25$, then $s_{(2,1)} = e_1 e_2 - e_3 \le 8/5$, with equality only at a permutation of $(4/5, 4/5, 1/5)$. To prove the lemma, sort $a \ge b \ge c$, put $u = a + b$, $v = ab$, and maximize first in $v$ under $v \le u^2/4$ and $v \le 24/25 - cu$. The bounds exchange at $c = 1/5$; after imposing either $e_1 = 9/5$ or $e_2 = 24/25$, subtraction from $8/5$ is respectively

$$\frac{(5c-1)(25c^2 + 50c - 71)}{500} \quad (0 \le c \le 1/5), \qquad \frac{(5c-4)^2(5c-1)}{125} \quad (1/5 \le c \le 3/5).$$

Both expressions are nonnegative on their stated intervals. Equality forces $c = 1/5, u = 8/5, v = 16/25$. The determinant equality case then forces a balanced Boolean control. □

We next vary the conference phase. Let $C = C^*$ be an arbitrary Hermitian conference matrix of order six, so $C_{ii} = 0$, $|C_{ij}| = 1$, and $C^2 = 5I$. For a balanced triple $T = \{i, j, k\}$, write

$$A = C[T], \qquad R = C[T, T^c], \qquad H_T = RR^*/5 = I - A^2/5,$$

and define the real triangle holonomy

$$r_T = \mathrm{Re}(C_{ij}C_{jk}C_{ki}).$$

For continuous controls choose orthonormal frames $Q_+, Q_-$ for the two eigenspaces of $C$, put $K_C(x) = Q_-^* D_x Q_+$, and apply the same symmetric-function notation to $K_C(x)^* K_C(x)$. The known order-six Hermitian conference family realizes every $r_T^2 \in [0,1]$: specialize Et-Taoui's displayed $C(a,b)$ to $k=3$ and $a=1$, for which $C(1,b)$ is Hermitian and a fixed triangle has $r_T = \mathrm{Re}\, b$ [13, Theorems 2 and 6]. The proofs do not use the classification of this family.

**Theorem 5.2** (Hermitian exchange landscape)**.** *For every balanced triple,*

$$p_1 = \frac{9}{5}, \qquad p_2 = \frac{33}{25}, \qquad e_2 = \frac{24}{25},$$

*while*

$$e_3 = \frac{4(5-r_T^2)}{125}, \qquad h_3 = \frac{317-4r_T^2}{125}, \qquad s_{(2,1)} = \frac{196+4r_T^2}{125}. \tag{4}$$

*For arbitrary real controls $x \in [-1,1]^6$, uniformly over this Hermitian conference class,*

$$p_1 \le \frac{9}{5}, \quad p_2 \le \frac{33}{25}, \quad e_2 \le \frac{24}{25}, \quad e_3 \le \frac{4}{25},$$
$$h_3 \le \frac{317}{125}, \quad s_{(2,1)} \le \frac{8}{5}.$$

*The complete componentwise-maximal frontier of $(h_3, s_{(2,1)}, e_3)$ is*

$$\frac{1}{125}(317-4t,\ 196+4t,\ 20-4t), \qquad 0 \le t \le 1. \tag{5}$$

*It is realized by balanced Boolean controls with $t = r_T^2$. Its $t=0$ endpoint maximizes the symmetric and exterior sectors, while $t=1$ maximizes the mixed sector.*

*Proof.* The triangle block has characteristic polynomial $z^3 - 3z - 2r_T$. Newton identities applied to $H_T = I - A^2/5$ give the fixed first two moments and determinant in (4); the identities $h_3 = e_3 + p_1 p_2$ and $s_{(2,1)} = p_1 e_2 - e_3$ give the other sectors.

For continuous controls, the convexity and rank-one-minor arguments in Theorem 5.1 do not use real conference entries through $e_2$. Separate convexity of $|\det K|^2$ reduces $e_3$ to Boolean cuts, where (4) gives $e_3 \le 4/25$. The same symmetric-function identities give the remaining bounds. In addition,

$$s_{(2,1)} + e_3 = p_1 e_2 \le \frac{216}{125}, \qquad h_3 + s_{(2,1)} = p_1(p_2 + e_2) \le \frac{513}{125}.$$

For a feasible point $(h,s,e)$, put

$$L = \max\left\{0, \frac{125s-196}{4}\right\}, \qquad U = \min\left\{1, \frac{317-125h}{4}, \frac{20-125e}{4}\right\}.$$

The individual bounds and the two supporting inequalities give $L \le U$. Every $t \in [L,U]$ therefore gives a point of (5) that dominates $(h,s,e)$. Conversely, the cited one-parameter family realizes every $t \in [0,1]$. Both supporting inequalities are equalities on the segment, so no two distinct points on it dominate one another. $\square$

Ordinary three-uniform ETF compressions retain the sign of $\operatorname{Re} z_T$ and select a purely imaginary/skew class [8, 9]. The exchange operator instead sees $(\operatorname{Re} z_T)^2$; its rigidity endpoint is real.

**Theorem 5.3** (Squared-spectrum rigidity and stability)**.** *For an order-six Hermitian conference matrix, the following are equivalent: the balanced squared cross-block spectrum is cut-independent; any one of $e_3, h_3, s_{(2,1)}$ is cut-independent; and $C$ is switching/permutation equivalent to a real symmetric conference matrix.*

*Complementary triples have the same squared real holonomy, so choose one triple from each of the ten complementary balanced-cut pairs and put*

$$\delta(C) = 1 - \frac{1}{10}\sum_T r_T^2.$$

*Then*

$$\overline{e}_3 = \frac{16+4\delta}{125}, \qquad \overline{h}_3 = \frac{313+4\delta}{125}, \qquad \overline{s}_{(2,1)} = \frac{200-4\delta}{125}.$$

*Let $d_{\mathbb{R}}(C)$ be the unnormalized Frobenius distance from $C$ to the diagonal-unitary switching and permutation orbit of the real order-six conference matrices. Globally,*

$$d_{\mathbb{R}}(C)^2 \geq \frac{10}{3}\delta(C). \tag{6}$$

*If*

$$\delta(C) < \frac{6-\sqrt{35}}{20}, \tag{7}$$

*then*

$$d_{\mathbb{R}}(C)^2 \leq 40\delta(C). \tag{8}$$

*The constants and threshold are not claimed to be sharp.*

*Proof.* The proof first turns sector constancy into constant absolute triangle holonomy. Pentagon parity excludes the interior holonomy range, and a Pfaffian square excludes the purely imaginary endpoint. Triangle-product estimates then give the global metric bound; dephasing and sign rounding give the local reverse bound.

The complementary cross blocks are $R$ and $R^*$, so they have the same squared singular spectrum and hence the same $r_T^2$. By (4), any one-sector constancy is equivalent to $|r_T| = \rho$ for every triangle. Dephase one row and column to ones and write the remaining five-by-five block as $S$. Root triangles force every off-diagonal entry to have real part $\pm\rho$. If $0 < \rho < 1$, write

$$S = \rho A + i\sqrt{1-\rho^2}\, B,$$

where the row sums make $A$ a symmetric sign matrix with two positive and two negative neighbors at each vertex and $B$ a regular skew tournament. The positive edges of $A$ form a pentagon; label them $01, 12, 23, 34, 40$. The imaginary part of $S^2 = 5I - J$ requires $AB + BA = 0$, but three entries satisfy

$$(AB+BA)_{02} + (AB+BA)_{03} - (AB+BA)_{23} = 2(b_{03} - b_{24} - b_{34}),$$

whose parenthesized sum of three signs cannot vanish. At $\rho = 0$, bordering $B$ by a row and column of signs gives an integral six-by-six skew matrix $\widetilde{B}$ with $\widetilde{B}^2 = -5I$. Its determinant would be 125, impossible because an even integral skew determinant is an integral Pfaffian square. Thus $\rho = 1$, the real class. The converse is immediate. Averaging (4) gives the three defect identities.

For stability, align $C$ with any real-orbit representative $R$. If $z_T$ and $w_T \in \{\pm 1\}$ are their triangle products, then

$$1-(\operatorname{Re} z_T)^2 \leq |z_T-w_T|^2 \leq 3\sum_{e\subset T}|C_e-R_e|^2.$$

Each edge belongs to four of the twenty triangles and complementary triples have equal squared real holonomy. Summing gives $20\delta \leq 6\|C-R\|_F^2$, proving (6).

For the reverse estimate, dephase at a root and round each of the ten other upper-triangular unit entries $u$ to a nearest sign $\sigma(u)\in\{\pm 1\}$, choosing either sign when $\operatorname{Re} u = 0$. This gives a real sign matrix $R$, and

$$|u-\sigma(u)|^2 = 2(1-|\operatorname{Re} u|) \leq 2(1-(\operatorname{Re} u)^2),$$

so the ten root triangles give

$$\|C-R\|_F^2 \leq 40\delta.$$

Put $x=\|C-R\|_F$. Since $C^2=5I$,

$$\|R^2-5I\|_2 \leq x(2\sqrt{5}+x).$$

Every off-diagonal entry of $R^2$ is an even integer and every diagonal entry is five. Under (7), $x<\sqrt{7}-\sqrt{5}$, so the displayed norm is less than two; parity forces $R^2=5I$. This rounded $R$ is therefore an exact real conference matrix and proves (8). □

# 6 Balanced controls and calibrated readout

With fixed physical port axes, a permanent is an ordinary transition amplitude, accessible only through a phase-sensitive multiphoton protocol; photon counting measures its squared modulus. Such a value belongs to the calibrated apparatus rather than to the unmarked Golden operator. Likewise, the determinant sign belongs to the oriented amplitude: the probability $F_{\text{ext}}$ does not retain it. The precursor below infers that sign from phase-referenced one-particle tomography; it does not measure a direct many-fermion phase.

The necessity of balance is independent of the Golden conference matrix.

**Proposition 6.1** (Universal balance obstruction)**.** *Let $V_+\oplus V_-$ be an orthogonal splitting of $\mathbb{R}^{2d}$ into two $d$-planes. For a Boolean path control with negative support $S$, the cross block $K_S : V_+\to V_-$ satisfies*

$$\operatorname{rank} K_S \leq \min\{|S|, 2d-|S|\}.$$

*Consequently a nonzero filled $d$-fermion amplitude is possible only for a balanced $d+d$ control.*

*Proof.* Writing $P_S$ for the coordinate projection onto the negative paths and using orthogonality of the two port spaces gives

$$K_S = -2Q_-^{\mathsf{T}}P_SQ_+,$$

so $\operatorname{rank} K_S \leq |S|$. Complementing the control negates its cross block and gives the second bound. □

There are 64 Boolean controls $x \in \{\pm 1\}^6$. Proposition 6.1 bounds the rank of the $2\binom{6}{0} + 2\binom{6}{1} + 2\binom{6}{2} = 44$ unbalanced masks by two. The common balanced spectrum makes all $\binom{6}{3} = 20$ remaining masks invertible. The balanced controls are therefore exactly those with nonzero determinant, and all attain the same Boolean optimum:

$$|\det K_T| = \frac{4}{5\sqrt{5}}, \qquad P_F = |\det K_T|^2 = \frac{16}{125}.$$

The pivot convention above fixes the port frames. In that gauge the collision-free bosonic probabilities on balanced controls are

$$P_B(111 \mid 111) \in \left\{ \frac{16}{3125}, \frac{36}{3125}, \frac{64}{3125} \right\}.$$

These are calibrated control values, not port-gauge invariants. Coherent access to the amplitudes distinguishes all twenty oriented masks; squaring pairs antipodal controls and loses the overall orientation.

Across the ten projective balanced cuts, the six determinant sign words have pairwise Hamming distance six. Three selected signs identify a marked protocol once Golden membership is already known. Five signs and magnitudes jointly test membership and identify the protocol. Measuring all ten signs supports soft matched-filter decoding; hard nearest-word decoding corrects two sign errors. No relative sign word fixes the global orientation without a coherent port reference.

For an explicit schedule, order the projective cuts lexicographically by the negative supports containing path zero:

$$012, 013, 014, 015, 023, 024, 025, 034, 035, 045.$$

The first three positions $012, 013, 014$ classify the six protocols after membership is known. The five-cycle $012, 015, 023, 034, 045$ jointly tests the common Golden magnitudes and classifies the protocol. These are fixed schedules for the transported marking above, not protocol-dependent choices.

Order the protocols by the one-line forms of their chosen path permutations. In the common calibrated orientation, the three-cut decoder is

| protocol | $T_0$ | $T_1$ | $T_2$ | $T_3$ | $T_4$ | $T_5$ |
|---|---|---|---|---|---|---|
| $P_T$ | 012345 | 012354 | 012435 | 012453 | 012534 | 012543 |
| signs | $---$ | $--+$ | $-++$ | $+--$ | $++-$ | $+++$ |

The supplemental certificate records all ten signs for every marked protocol. Reversing the common orientation negates every signature; it does not change the relative classifier once that orientation is calibrated.

The universal complementarity identities for determinants and permanents of Jabbour and Cerf [6] provide a useful calibrated consistency check. They do not distinguish the Golden family by themselves.

# 7 Photonic design limit

Figure 2 separates three experiments that use the same one-particle transfer but answer different questions. Coherent light recovers a phase-referenced matrix and hence an inferred determinant sign. Ordinary three-photon input measures calibrated bosonic probabilities. Direct three-fermion emulation instead requires three identical copies of the transfer and an antisymmetric three-qutrit resource. The first two form a feasible precursor; the third is conditional on the additional source.

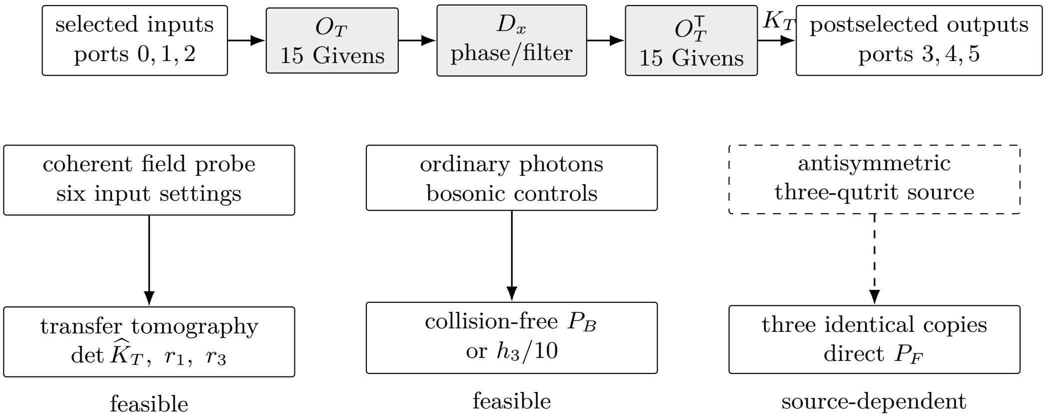


Figure 2: Circuit and readout boundary. The upper line implements $U_T(x) = O_T^\mathsf{T} D_x O_T$ in the filtered realization; a balanced mask may instead use the direct fifteen-cell compilation. The indicated subblock maps input ports $0, 1, 2$ to output ports $3, 4, 5$. Solid arrows are demonstrated measurement primitives. The dashed box and arrow mark the additional resource needed for direct three-fermion emulation. The structural nulls are $r_1 = \sum_T \widehat{Z}_T$ and $r_3 = \sum_T \widehat{Z}_T^3$; the drawing uses no color coding.

## Circuit, calibration, and error bounds

Put $O_T = [Q_{T,+}\ Q_{T,-}]$, inject eigenports $0, 1, 2$, and postselect outputs $3, 4, 5$. Thus $K_T$ is the corresponding off-diagonal block of $U_T = O_T^\mathsf{T} D_x O_T$. An adjacent-mode Givens decomposition uses fifteen cells for $O_T$; the full-precision netlist and the rounded six-decimal table are retained in the supplement, with maximum reconstruction errors below $2 \times 10^{-14}$ and $7 \times 10^{-7}$, respectively. Balanced orthogonal controls admit a direct fifteen-cell compilation plus one fixed sign flip. A general real filter uses two meshes and six variable attenuators, with amplitude transmission $x_i$ and intensity transmission $x_i^2$. The other marked protocols are obtained by the chosen path permutations.

Injecting each input coherently and measuring every output against one common local oscillator fixes the phase-referenced transfer in six input settings. The common output reference is essential: intensity data and input-only interference leave independent output-row phases free and cannot determine a block-determinant sign. The reconstruction primitive is demonstrated experimentally [25]; the common input–output phase calibration is an additional assumption here. Programmable twelve-mode processors have separately operated with three photons [26].

Before comparing probabilities, reconstruct the six marked amplitudes in one gain and orientation convention and test the structural nulls

$$r_1 = \sum_T \widehat{Z}_T, \qquad r_3 = \sum_T \widehat{Z}_T^3.$$

Independent normalization of the protocols would erase these identities. Diagonal loss remains within the controlled family, whereas mesh error can move the amplitudes off the common Golden carrier. Passing both nulls is necessary but not sufficient; the controlled exchange spectra and calibrated probabilities provide the subsequent tests.

For two contraction blocks $K$ and $K + E$,

$$|\det(K + E) - \det K| \le 3\|E\|_2. \tag{9}$$

Indeed, integrate the derivative along $K + tE$: the trace norm of its adjugate is at most three because its singular values are pairwise products of contraction singular values. At a balanced real control the determinant sign is therefore certified by

$$\|E\|_2 < \frac{4}{15\sqrt{5}}.$$

The supplemental package retains the exact decoder, compilation, accepted- trial formulas, and source-quality gates. They are conditional design calculations rather than theorem inputs or claims of experimental performance.

### External resource dependency

The emulator of Matthews et al. [21] sends an entangled resource through identical copies of a one-particle process. Direct three-fermion emulation here requires the totally antisymmetric state of three photonic qutrits. Goyal et al. [22] and Piccolini et al. [23] give preparation proposals. No preparation and characterization experiment was located in the bounded audit, so the source remains an external dependency rather than a component claimed here. A demonstrated three-qutrit GHZ state [24] has different exchange symmetry.

The value $P_F = \det H$ assumes normalized antisymmetric input and output states. The six permutation amplitudes are combined coherently in one normalized antisymmetric projection, so no factor of 3! multiplies the stated probability. Resource fidelity, photon indistinguishability, matching of the three transfer copies, and analyzer calibration remain separate requirements. The supplement records conservative mixture- and fidelity-based quality gates; without a characterized noise model they are worst-case design bounds, not performance predictions.

## 8 Discussion

The Golden transfer separates three levels of information. Singular values determine every intrinsic exchange scalar; orienting the determinant lines adds the sign of the top exterior amplitude; phase-referenced port frames then expose calibrated amplitudes such as the permanent. Antisymmetric exchange realizes the determinant amplitude, but its event probability discards the sign. Coherent one-particle tomography can infer that sign only after the laboratory frames have been compiled and oriented.

The exchange landscape has two independent deformations. Continuous real control moves inside a fixed Golden operator and has the balanced Boolean points as its unique degree-three joint optimum. Hermitian conference phase moves the balanced endpoint itself: the first two moments remain fixed while triangle holonomy transfers weight along the exact Pareto segment. Forgetting the holonomy sign changes the rigidity class from the purely imaginary class of ordinary three-uniform spectra to the real conference class of squared exchange spectra. The averaged defect both records the mean sector transfer and provides a quantitative local coordinate transverse to the real orbit.

The order-six real point remains exceptional in the order direction as well: it is the only nontrivial symmetric conference order with a cut-independent balanced spectrum. Higher orders retain exact mean and variance but not a universal spectrum. Finally, the optical section separates a feasible one-particle and bosonic certification from direct three-fermion emulation, which still depends on an additional antisymmetric resource. These are mathematical and design boundaries, not a claim of a built device.

## Data availability

No empirical data were generated or analyzed. The exact mathematical artifacts supporting the finite claims—the standard-library checker, compact certificate, SHA-256 manifest, locked symbolic environment, and independent replay entry points—are included with the article's supplemental package. The submitted artifact version is identified in that package; live hosting and its public locator are supplied with the publication record.

## Verification

Human proofs cover the orbit theorem, continuous-control reduction, Hermitian holonomy formulas, exact Pareto edge, squared-spectrum rigidity, averaged defect, local metric stability, Boolean rank boundary, and decoder radius. Classical citations supply the higher-order design statistics; Paper III supplies labelled reconstruction. The local certificate independently checks the Boolean endpoint profiles, exchange identities, calibrated permanent values, six sign words, full-precision netlist, and fifteen-cell compilation. Three atomic source bundles provide exact generators and independent replays; one paper-local manifest pins every load-bearing file.

From the supplemental-package directory, `make check` verifies source hashes and independently recomputes every imported rational identity, census total, Hamming distance, schedule, and source replay used above. This evidence does not certify experimental feasibility, source availability, or literature priority.

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